

Damage-based failure theory and its application to 2D-C/SiC composites



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ABSTRACT

Damage evolution is of great importance to determine the final failure of ceramic matrix composites (CMCs). In order to characterize the damage coupling behavior and its effect on failure strength of CMCs, a new methodology for damage coupling analysis was formulated. And, new kind of damage-based failure criteria were established under plane stress state, including maximum damage criterion and quadratic damage criterion. Both the criteria require investigation upon the damage evolution laws. The proposed failure theory was applied to predict the coupled damage behavior and the off-axis strength of a 2D C/SiC composite. The theoretical results agree well with the experimental data.

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1. Introduction

Continuous fiber reinforced ceramic matrix composites usually present nonlinear mechanical behaviors [1], manifesting the interior damages such as matrix cracking [2–4], interfacial debonding [5–6] and fiber breakage [7]. Generally, damage is highly anisotropic in nature and is accompanied by an anisotropic degradation of the elastic properties of the material [8]. Moreover, when complex stress is applied to the material, damages in different material axes may interact with each other, complicating the stress–strain response and damage evolutions [4,9].

More and more works are devoted to investigate the damage behavior of CMCs, and many damage mechanics models have been developed to simulate the constitutive relationships [10–18]. In those models, damage coupling is successfully characterized by introducing thermodynamic forces as variables of the damage evolution laws. However, the final failure of the composites is rarely predicted [10–11]. Actually, it is promising to predict material strength by constructing damage-based failure criterion, since the damage evolution laws have already been obtained. Nevertheless, only a few damage-based failure conditions had appeared in the past years [11,18–21]. And, they are impossible to form a unified theoretical system to facilitate applications.

Up to now, the widely used failure theories are still stress-based, strain-based and energy-based [22–29]. Some of the theories have taken material nonlinearity into account. But, those models did not originate from the damage conception, though

damage evolution is significant for predicting the constitutive relation and final failure of the damageable composites [11,19]. Anyhow, the influence of the micro-damages and damage evolutions upon mechanical failure has not been adequately investigated.

In this study, attention is focused on the development of a strength theory in the framework of continuum damage mechanics under plane stress state. This work relies on the good understanding of the damage and failure mechanisms of CMCs. Since the final rupture of the advanced ceramic composites is actually driven by microstructural damage accumulation, a kind of damage-based failure theory is established to forecast the rupture strength. The advantage of this theory is the direct inclusion of damage evolution laws into the strength model, providing convenience for nonlinear analysis and strength design of materials and structures.

2. Theory description

2.1. Damage characterization

In order to describe the anisotropic damage evolution behavior and the damage coupling mechanisms of the nonlinear ceramic matrix composites, a unique damage matrix with nine damage components is introduced based on a decoupling conception under plane stress state (compressive stress is excluded), and its description is as below:

$$[D_{ij}] = \begin{bmatrix} D_{11} & D_{12} & D_{16} \\ D_{21} & D_{22} & D_{26} \\ D_{61} & D_{62} & D_{66} \end{bmatrix}, (i, j = 1, 2, 6) \quad (1)$$

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where D_{ij} denotes damage in principal material direction i induced by stress component σ_j . The six off-diagonal terms are used to describe the damage coupling effects. Considering material stiffness change, the damage components can be defined as

$$D_{ij} = \frac{\Delta E_i(\sigma_j)}{E_i}, (i, j = 1, 2, 6) \quad (2)$$

where E_i is the original engineering elastic constant, while $\Delta E_i(\sigma_j)$ is the change of E_i caused by the applied stress σ_j . Essentially, coupling term D_{ij} ($i \neq j$) reflects the contribution of the principal damage D_{ij} induced by stress σ_j to the variation of the stiffness E_i . This means that, for example, shear damage D_{66} resulted from shear stress σ_6 simultaneously influence other elastic coefficients, not only the shear modulus E_6 . Generally, the damage matrix is not symmetric, and it is consumed to be completely quantified.

Many experimental and theoretical studies reveal that damage propagation would become much complicated under complex stress states due to damage coupling effects [10–14]. For instance, damage evolution in 1-axis of the material is not only dominated by D_{11} but also affected by D_{22} and D_{66} . Therefore, the effective damage variables d_i ($i = 1, 2, 6$) must be introduced to characterize such mechanical features. The scalar variables d_i ($i = 1, 2, 6$) are usually defined as [10,19]:

$$d_1 = 1 - \frac{E_{1,d}}{E_1}, \quad d_2 = 1 - \frac{E_{2,d}}{E_2}, \quad d_6 = 1 - \frac{E_{6,d}}{E_6} \quad (3)$$

where $E_{1,d}$, $E_{2,d}$ and $E_{6,d}$ are the damaged moduli corresponding to E_1 , E_2 and E_6 , respectively. Based on the physical definitions of the damage components listed in the matrix, the effective damage variables can be approximated by

$$\begin{cases} d_1 = D_{11} + D_{12} + D_{16} \\ d_2 = D_{21} + D_{22} + D_{26} \\ d_6 = D_{61} + D_{62} + D_{66} \end{cases} \quad (4)$$

It is noted that d_i ($i = 1, 2, 6$) are now formulated as complicated variables, which have incorporated the damage coupling effects. From the other perspective, Eq. (4) can also be replaced by the following equation through introducing related coupling coefficients [11] for the coupling terms given in the damage matrix, as

$$\begin{Bmatrix} d_1 \\ d_2 \\ d_6 \end{Bmatrix} = \begin{bmatrix} \omega_{11} & \omega_{12}(\sigma_2) & \omega_{16}(\sigma_6) \\ \omega_{21}(\sigma_1) & \omega_{22} & \omega_{26}(\sigma_6) \\ \omega_{61}(\sigma_1) & \omega_{62}(\sigma_2) & \omega_{66} \end{bmatrix} \begin{Bmatrix} D_{11} \\ D_{22} \\ D_{66} \end{Bmatrix} \quad (5)$$

where $\omega_{11} = \omega_{22} = \omega_{66} = 1$, while other coupling coefficients are defined as functions of the associated stress components.

Especially for plain-woven CMC, we have $\omega_{12} = \omega_{21}$, $\omega_{16} = \omega_{26}$, $\omega_{61} = \omega_{62}$ for structural symmetry. Subsequently, Eq. (5) is reduced to

$$\begin{Bmatrix} d_1 \\ d_2 \\ d_6 \end{Bmatrix} = \begin{bmatrix} 1 & \omega_{12}(\sigma_2) & \omega_{16}(\sigma_6) \\ \omega_{12}(\sigma_1) & 1 & \omega_{16}(\sigma_6) \\ \omega_{61}(\sigma_1) & \omega_{61}(\sigma_2) & 1 \end{bmatrix} \begin{Bmatrix} D_{11} \\ D_{22} \\ D_{66} \end{Bmatrix} \quad (6)$$

In Eq. (5), D_{11} and D_{22} can be obtained by on-axis tests, while D_{66} can be measured by pure shear test. To determine a coupling coefficient by a decoupling experiment, taking ω_{61} for example, one can shear a coupon with axial preloading history [30,31]. Similar methods can be designed to acquire the remained coupling coefficients. Moreover, biaxial loadings and tension–shear combined loadings [32,33] are also available for coupling research. Anyhow, the key problem remained is to identify the damage evolution laws on the basis of experimental measurement and/or theoretical analysis.

In Eqs. (4)–(6), it is concluded that a positive coupling term acts to decrease the material strength by enhancing its interior damage, while a negative one plays the opposite role. However, whether the coupling effect is positive or negative requires being determined according to the material attribute and the imposed loading condition in practical problems.

2.2. Maximum damage criterion

In general, the failure of a ceramic matrix composite is relevant to its interior damage. When mechanical fracture happens, the damage components must satisfy a mathematical condition. A simple assumption is that when the effective damage in any principal material direction exceeds the damage value associated to the failure, the mechanical rupture occurs. This failure assumption can be stated as:

$$\begin{cases} d_1 = D_{11,\max}, & \text{or} \\ d_2 = D_{22,\max}, & \text{or} \\ d_6 = D_{66,\max}, \end{cases} \quad (7)$$

where $D_{ij,\max}$ ($j = 1, 2, 6$) are the critical values of the principal damage variables. Note that Eq. (7) is the first proposal of this paper. Although this failure criterion resembles the maximum stress/strain theory in style, it makes much progress in considering the interactions among the stress components by including the damage coupling effects. Thus, Eq. (7) can be expected to reach better accuracy than the maximum stress criterion in macro-scale for its sacrifice of simplicity.

2.3. Quadratic damage criterion

In order to overcome the complexity of the maximum damage criterion, all of the damage coupling terms are expected to be excluded from the theoretical equation of the established criterion. In other words, the failure model only involves the principal damage components such as D_{11} , D_{22} and D_{66} . Under such consideration, it is assumed that there exists a failure surface in the damage-space in the following form:

$$I_1 D_{11} + I_2 D_{22} + I_6 D_{66} + I_{11} D_{11}^2 + I_{22} D_{22}^2 + I_{66} D_{66}^2 + I_{12} D_{11} D_{22} + I_{16} D_{11} D_{66} + I_{26} D_{22} D_{66} = 1 \quad (8)$$

The above equation is the second proposal of this paper, which is similar in style to the stress-based Tsai–Wu criterion [24]. It is concluded that damage interaction effects are naturally included in the quadratic model. Further, a single expression isolated from the model will suggest a single damage-based failure condition. This means that, when Eq. (8) is considered for use, the irrelevant parameters should be eliminated first according to the specific coupling mechanisms of the material to simplify the model.

As typical cases, the following failure conditions are isolated from Eq. (8):

$$I_{11} D_{11}^2 + I_{22} D_{22}^2 + I_{66} D_{66}^2 = 1 \quad (9a)$$

$$I_{11} D_{11}^2 + I_{12} D_{11} D_{22} + I_{22} D_{22}^2 + I_{66} D_{66}^2 = 1 \quad (9b)$$

$$I_1 D_{11} + I_{12} D_{11} D_{22} + I_2 D_{22} + I_{66} D_{66}^2 = 1 \quad (9c)$$

In Eq. (9), I_1 , I_2 , I_{11} , I_{22} and I_{66} are simply set as invariables, so that they can be determined by uniaxial tension and in-plane shear experiments. If one impose a uniaxial tensile stress oriented along the 1-axis or 2-axis on a specimen, then the failure condition is $D_{11} = D_{11,\max}$ or $D_{22} = D_{22,\max}$. Similarly, by imposing pure in-plane shear stress on a sample, the failure condition becomes

$D_{66} = D_{66,\max}$. Eqs. (9a)–(9c) must apply to such single loading cases, respectively. Therefore, we readily obtain the following solutions:

$$\left(\frac{D_{11}}{D_{11,\max}}\right)^2 + \left(\frac{D_{22}}{D_{22,\max}}\right)^2 + \left(\frac{D_{66}}{D_{66,\max}}\right)^2 = 1 \quad (10a)$$

$$\left(\frac{D_{11}}{D_{11,\max}}\right)^2 + \left(\frac{D_{22}}{D_{22,\max}}\right)^2 + \left(\frac{D_{66}}{D_{66,\max}}\right)^2 + I_{12}D_{11}D_{22} = 1 \quad (10b)$$

$$\left(\frac{D_{66}}{D_{66,\max}}\right)^2 + \frac{D_{11}}{D_{11,\max}} + \frac{D_{22}}{D_{22,\max}} + I_{12}D_{11}D_{22} = 1 \quad (10c)$$

In Eq. (10), I_{12} is an unknown parameter modifying the coupling intensity between the two associated damage components and thus allowing for data fitting. Eqs. (10b) and (10c) are similar in style to the stress-based Tsai–Hill criterion and Yeh–Stratton criterion, respectively [22,26].

Eqs. (10a)–(10c) are the final three important proposals of this paper. For a certain material system, it is expected that at least one of the newly proposed failure conditions could be appropriate. If one only intends to obtain the failure strength, using Eqs. (8) and (10) would be more efficient than Eq. (7). It is emphasized that Eq. (10) only presents some confined damage interaction patterns. For other different damage coupling mechanisms, Eq. (8) should be reconsidered for use.

It is also noted that there is no difference between the quadratic damage criterion and the maximum damage criterion in determination of the damage evolution laws. For principal damage components such as D_{11} , D_{22} and D_{66} , the damage functions (i.e., $D_{11}(\sigma_1)$, $D_{22}(\sigma_2)$ and $D_{66}(\sigma_6)$) must be measured by on-axis tension and in-plane shear experiments.

At this point, a new methodology named *D failure theory* is developed for advanced ceramic composites with nonlinear mechanical behavior, which consists of a methodology for characterizing the anisotropic damage evolution laws and several damage-based failure criteria for rupture evaluation. Eqs. (7), (8) and (10) are all available for advanced ceramic composites. Certainly, this theory is also recommended for other nonlinear materials. Since the definition of the damage variables is usually from macroscopic consideration, this strength theory is thus appropriate for structural failure analysis in the framework of continuum damage mechanics.

3. Experiment and results

3.1. Material and tests

The material used in this investigation was a cross-woven C/SiC composite fabricated by isothermal CVI technique [34]. The fiber volume fraction of the preform was 40% and the bulk density of as-fabricated material was about $2.0 \text{ g}\cdot\text{cm}^{-3}$. The fiber/matrix interface was designed relatively strong. Tension and shear specimens were cut from a composite panel and the geometries are schematically shown in Fig. 1. Strain gauges were mounted on each coupon (see Fig. 1) to monitor the deformation. On-axis tension, off-axis tension and in-plane shear experiments were conducted, respectively. All the tests were performed on a Instron 1196 machine with a 100 kN load cell under displacement control at a crosshead speed of 0.5 mm/min.

3.2. Results and discussion

Typical on-axis tensile loops are schematically shown in Fig. 2, where E and E_u are the initial modulus and unloading modulus, respectively. It can be seen that the hysteresis loops are somewhat narrow and well-defined. The unloading modulus decreases with

increasing of unloading stress. Consequently, E_u is virtually the post-damage modulus. Fig. 3 exhibits the evolution of E_u with unloading stress. The tendency of this curve is not linear but parabolic.

Typical in-plane shear behavior of the 2D C/SiC composite is presented in Fig. 4, where G_u is the unloading shear modulus. The distinct nonlinear curve implies that micro-damages are continuously accumulated upon loading. Especially, the reloading portion below previous unloading shear stress has transformed into a nearly linear curve. Therefore, G_u is also a post-damage modulus, which can reflect the shear damage degree. Fig. 5 shows the variation of G_u with unloading shear stress and a linear relationship can be seen, which is referred to as linear damage.

The off-axis tensile behavior of the material is typically depicted in Fig. 6. The off-axis angles between fiber orientation and loading directions (see Fig. 1) are 15° , 30° and 45° , respectively. With increasing bias angle, significant variation of the stress/strain curves as well as the modulus and strength can be observed for the 2D-C/SiC composite. This phenomenon implies the influence of damage coupling on mechanical response of the material. In order to investigate the evolution behavior of the effective damage variable d_1 , the stress/strain response in 1-axis of the material (i.e., σ_1/ε_1 curve) at each bias angle was measured, as is shown in Fig. 7. Similar to the on-axis experiment, the unloading modulus defined by the hysteresis loop of σ_1/ε_1 curve can reflect the damage degree in 1-axis of the material under off-axis load.

The basic on-axis mechanical properties of the as-received composite are listed in Table 1. Meanwhile, the average off-axis modulus, Poisson's ratio and strength of the specimens under each bias loading case are given in Table 2. Such experimental data will be crucial for further damage evolution and failure strength predictions.

4. Theory application

4.1. On-axis damage identification

The unloading modulus is reasonable for principal damage definition according to the damage features of the 2D C/SiC composite, which is

$$D_{11} = \frac{E - E_{1u}}{E}, D_{22} = \frac{E - E_{2u}}{E}, D_{66} = \frac{G - G_u}{G} \quad (11)$$

where E and G are the Young's modulus and shear modulus, respectively. E_{1u} , E_{2u} and G_u are the unloading moduli in the material axes. According to the on-axis damage characteristic described in Section 3.1, one can get the following damage function for the material:

$$D_{ij} = A\sigma_j - B\sigma_j^2 \quad (12)$$

where $j = 1$ or 2 for structural symmetry. By fitting the curve involving unloading modulus and stress shown in Fig. 3, we obtain $A = 3.6 \times 10^{-3}$, $B = 5.7 \times 10^{-3}$. The evolution curve of the damage variable D_{ij} is also exhibited in Fig. 3. By substituting the maximum value of σ_j , i.e. tensile strength in Table 1, into Eq. (12), the parameter $D_{ij,\max}$ can be obtained as 0.5224. On the other hand, the linear curve involving shear damage and shear stress appears in Fig. 5 can be formulated as

$$D_{66} = A_{66}\sigma_6/G = B_{66}\sigma_6 \quad (13)$$

where both A_{66} and B_{66} are unknown parameters to be determined. Through fitting of experimental data, we obtain $B_{66} = 4.3 \times 10^{-3} \text{ MPa}^{-1}$, $G = 37.53 \text{ GPa}$, and $A_{66} = 0.16$. Corresponding to the shear strength, the critical value of $D_{66,\max}$ is about 0.5405.

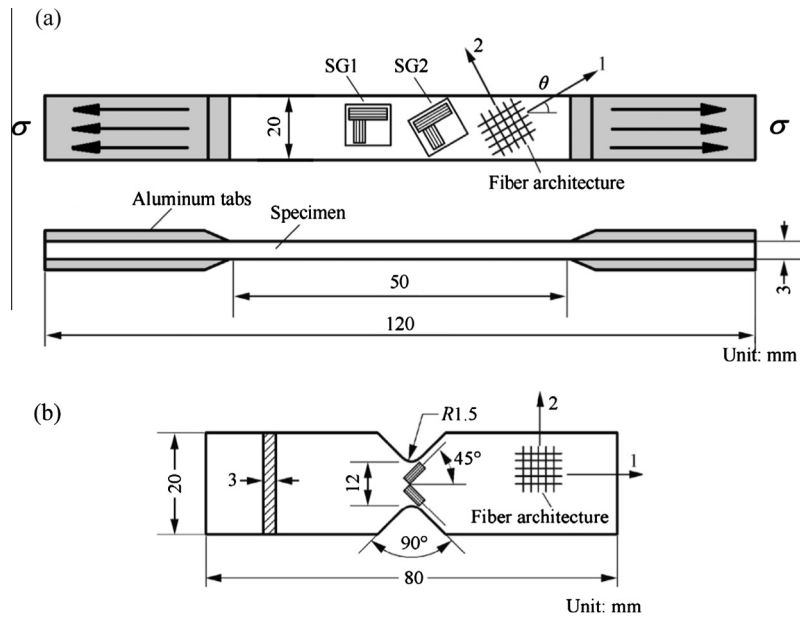


Fig. 1. Geometries of the tension and in-plane shear specimens. Two oriented strain gage rosettes (SG1 and SG2) are used for tensile sample; while a two gage $\pm 45^\circ$ rosette is used for shear sample.

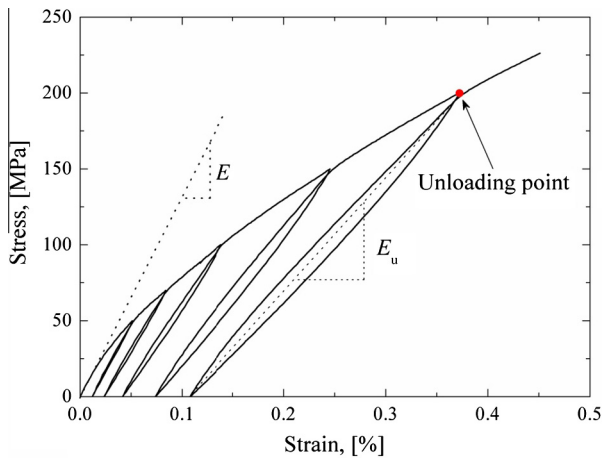


Fig. 2. Typical on-axis tensile behavior of 2D-C/SiC composite. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

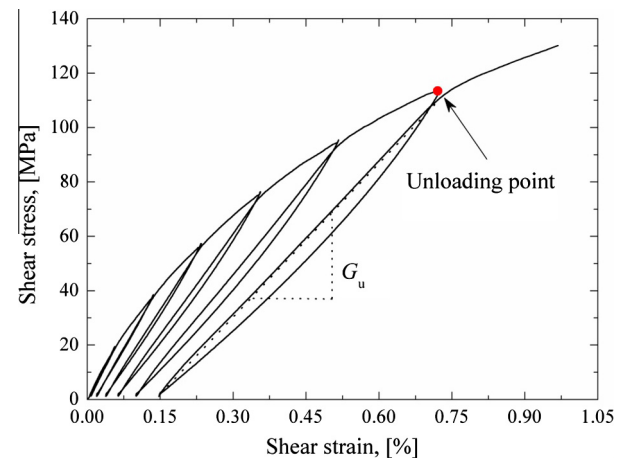


Fig. 4. Typical pure shear behavior of 2D-C/SiC composite. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

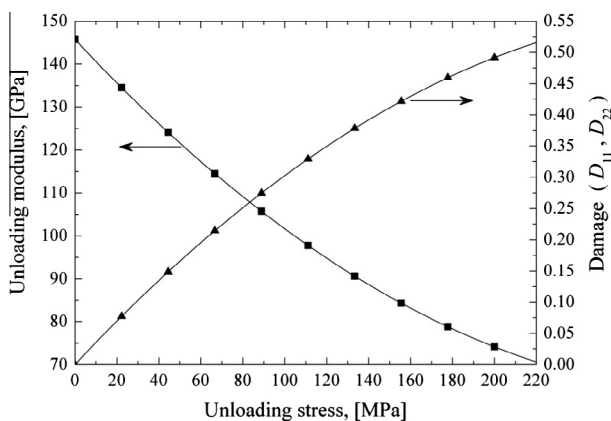


Fig. 3. On-axis damage evolution of 2D-C/SiC composite.

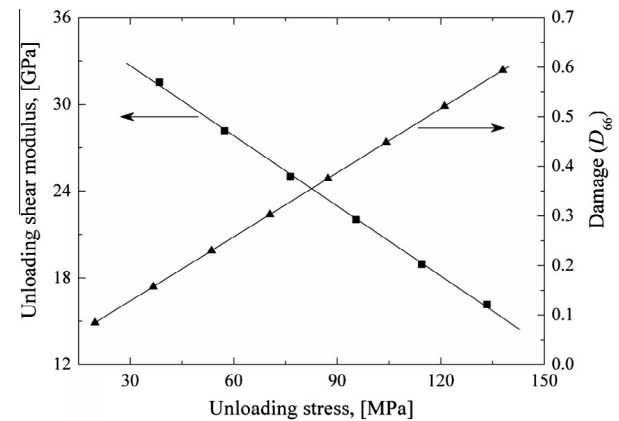


Fig. 5. In-plane shear damage evolution of 2D-C/SiC composite.

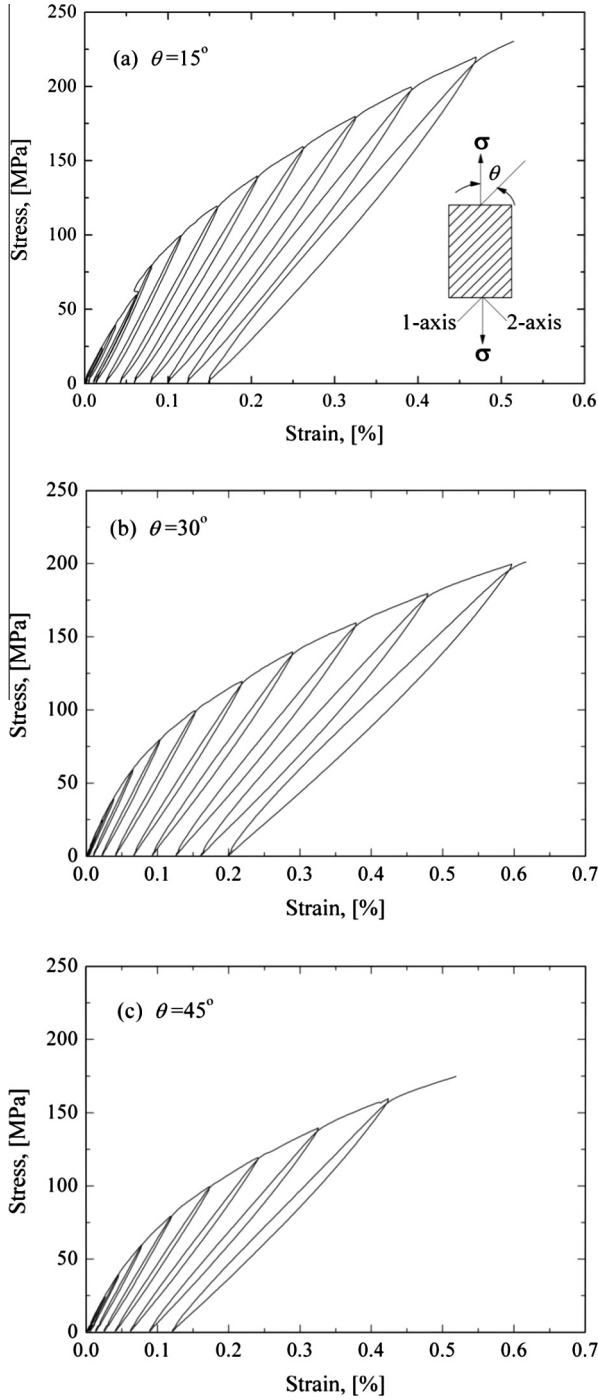


Fig. 6. Off-axis tensile behavior of 2D-C/SiC composite.

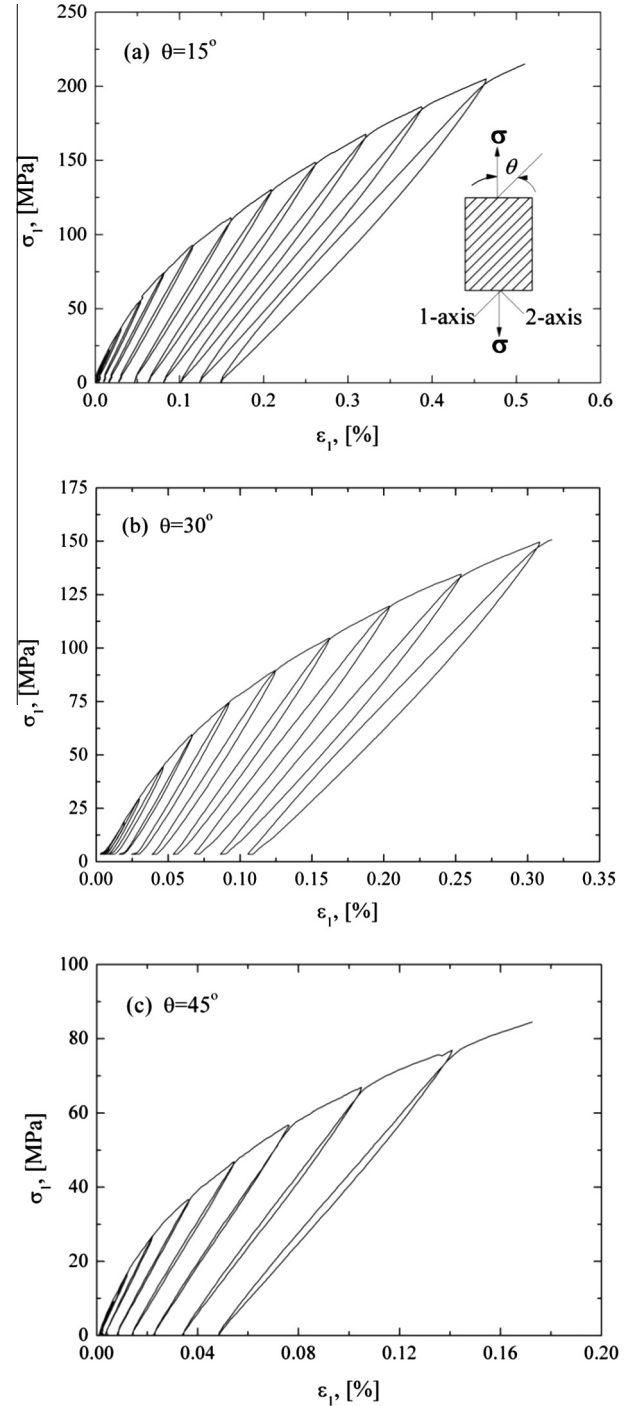


Fig. 7. Strain responses in 1-axis of the material under off-axis loadings.

4.2. Off-axis damage identification

When the composite material is subjected to off-axis load σ at angle θ to the fiber direction, the stress components in the principal material directions can be written as below:

$$\sigma_1 = \sigma \cos^2 \theta, \sigma_2 = \sigma \sin^2 \theta, \sigma_6 = \sigma \sin \theta \cos \theta \quad (14)$$

Furthermore, the effective damage variable d_1 can be determined experimentally by Eq. (3), where $E_{1,d}$ can be replaced by E_{1u} according to the previous analysis. The remained d_2 and d_6 are failed to be measured in this experiment, thus have to be identified.

Firstly, we introduced an off-axis damage variable:

$$d_\theta = (E_\theta - E_u^\theta)/E_\theta \quad (15)$$

where E_θ is the initial off-axis modulus, while E_u^θ is the associated unloading modulus. The experimentally measured curves of d_θ ($\theta = 15^\circ, 30^\circ$ and 45°) with σ are all depicted in Fig. 8. On the other hand, E_θ can also be theoretically estimated by the following equation:

$$\frac{1}{E_\theta} = \frac{\cos^4 \theta}{E_1} + \left(\frac{1}{E_6} - \frac{2\nu_{12}}{E_1} \right) \sin^2 \theta \cos^2 \theta + \frac{\sin^4 \theta}{E_2} \quad (16)$$

Table 1

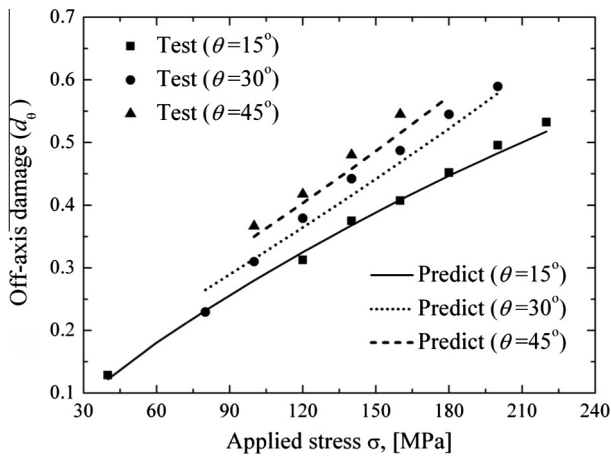
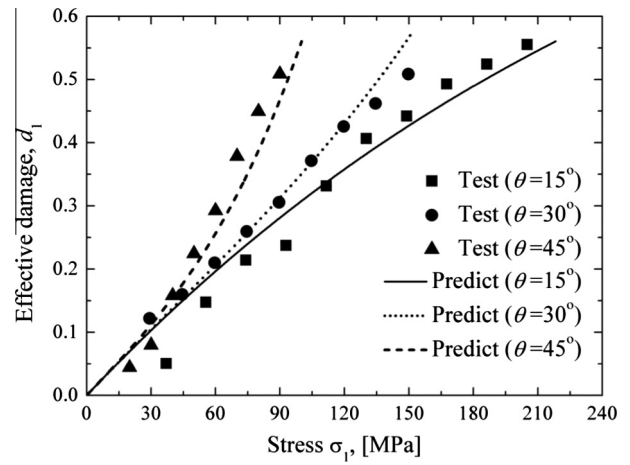
Mechanical properties of the 2D-C/SiC composite.

Material	Initial modulus (GPa)	Tensile strength (MPa)	Shear modulus (GPa)	Shear strength (MPa)	Poisson's ratio
2D C/SiC	145.7	226.0	37.5	125.7	0.07

Table 2

Off-axis experimental data and strength predictions of the 2D-C/SiC composite.

θ (°)	Experimental data			Strength predictions (MPa)			
	σ_u (MPa)	ν_{12}	E_θ (GPa)	Eq. (7)	Eq. (10a)	Eq. (10b)	Eq. (10c)
15	233.7 ± 3.4	0.2 ± 0.03	138.5 ± 5.3	211.4	202.1	216.7	209.1
30	200.6 ± 0.5	0.3 ± 0.01	125.7 ± 7.4	187.7	179.8	206.7	195.4
45	195.0 ± 20	0.3 ± 0.05	110.0 ± 4.1	175.5	172.7	203.3	191.8

**Fig. 8.** Off-axis damage evolutions of 2D-C/SiC composite.**Fig. 9.** Evolutions of effective damage in 1-axis of the material under off-axis loadings.

where ν_{12} is the Poisson's ratio at angle θ accounting for the micro-rotating of fiber bundles. In the same manner, E_u^θ can be evaluated by

$$\frac{1}{E_u^\theta} = \frac{\cos^4 \theta}{E_{1,d}} + \left(\frac{1}{E_{6,d}} - \frac{2\nu_{12}}{E_{1,d}} \right) \sin^2 \theta \cos^2 \theta + \frac{\sin^4 \theta}{E_{2,d}} \quad (17)$$

Once the damage coupling effect is considered, the damaged moduli $E_{1,d}$, $E_{2,d}$ and $E_{6,d}$ can be derived from the following equations:

$$\begin{cases} E_{1,d} = (1 - d_1)E_1 \\ E_{2,d} = (1 - d_2)E_2 \\ E_{6,d} = (1 - d_6)E_6 \end{cases} \quad (18)$$

where d_i ($i = 1, 2, 6$) can be calculated by Eq. (6). Based on data fitting conception, the coupling coefficients in Eq. (6) are simply provided as

$$\begin{aligned} \omega_{12}(\sigma_1) &= 1 \times 10^{-3} \cdot \sigma_1 \\ \omega_{12}(\sigma_2) &= 1 \times 10^{-3} \cdot \sigma_2 \\ \omega_{16}(\sigma_6) &= 5.26 \times 10^{-5} \cdot \sigma_6^2 \\ \omega_{61}(\sigma_1) &= \omega_{61}(\sigma_2) = 0.3 \end{aligned} \quad (19)$$

Combination of Eqs. (15)–(19) gives the off-axis damages d_θ ($\theta = 15^\circ, 30^\circ$ and 45°), which are also depicted in Fig. 8. It is seen that the predictions are in good agreement with the tested data. Consequently, the effective damage variables can be determined by Eq. (6) together with Eqs. (12), (13) and (19). Among d_i ($i = 1, 2, 6$), theoretical evolutions of d_1 with σ_1 at different angles are shown

in Fig. 9. A comparison of the simulated curves with the tested data reveals that the proposed method for damage coupling analysis is acceptable.

4.3. Off-axis tensile strength prediction

Verification upon the damage-based failure criteria is performed in this section with 2D C/SiC composites. The validation aims at Eqs. (7) and (10) because they are the basic formulations, while the applicability of the generalized damage criterion of Eq. (8) cannot be comprehensively examined here for lack of experimental data.

Since the damage evolution laws have already been obtained, the maximum damage criterion can be directly used to predict the off-axis tensile strength of the 2D-C/SiC composite. Substituting the critical values of the principal damage components into Eq. (7), the failure condition becomes

$$\begin{cases} d_1 = 0.5224, & \text{or} \\ d_2 = 0.5224, & \text{or} \\ d_6 = 0.5405 \end{cases} \quad (20)$$

The calculated results are listed in Table 2. It is seen that the predictions are slightly lower than the average tested data, implying the difference between the maximum damage criterion and the maximum stress criterion which usually gives much higher predictions than experimental data.

The following examination is placed on the predictive capability of Eq. (10). By substituting the principal damage evolution laws, i.e. Eqs. (12) and (13), into each theoretical equations (Eqs. (10a)–(10c)), we derive three groups of strength data, all of which are exhibited in Table 2. Note that for Eq. (10b), we set $I_{12} = -3.66$; while for Eq. (10c), we set $I_{12} = -8.2$. It can be inferred from Table 2 that each of the specified failure conditions can give acceptable strength predictions.

The previous comparative analysis demonstrates the reasonability and reliability of the newly proposed failure theory. In view that the associated stress components are variables of the damage evolution functions, the suggested damage-based failure criteria therefore have essential difference from the classical stress-based strength theories. *D* failure theory would be applicable in most macroscopic cases even when the target material resists to high temperature and complex environments. In such cases, accurate damage characterization is the precondition for material strength and life time predictions.

5. Conclusions

Based on the damage coupling mechanisms of CMCs, a new strength model named *D* failure theory is established in the framework of continuum damage mechanics. The theory consists of two parts: the methodology for damage coupling analysis and the damage-based failure criteria. The preliminary validation of the theory is performed on a 2D-C/SiC composite under off-axis tension. The experimental and theoretical results demonstrate that:

- The suggested *method for damage coupling analysis* is reasonable for highly anisotropic ceramic composites;
- The newly proposed *D* failure criterion has the potential to predict the off-axis strength of the advanced ceramic composites;
- Reasonable solutions of damage evolution laws are significant for accurate strength prediction by *D* failure criterion.

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